

STL analysis of Waterville monitoring data – Part III: Figures for the manuscript

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Introduction

STL analysis of long-term nutrient monitoring data from Heidelberg University. Data retrieved from National Center for Water Quality Research: <https://ncwqr.org/> and processed in Part I. Two STL functions are in Part II

Front matters

Setting up proper file paths, defining some useful functions, and loading necessary packages

Summarizing P and N loadings to WLE

```
wvl.daily <- cbind(wvl.daily, water_year(wvl.daily$date))
  ## Water year annual loading and spring loading
  ## annual
TPloadings <- tapply(wvl.daily$TPLoad, wvl.daily$water_year, sum)
SRPloadings <- tapply(wvl.daily$SRPLoad, wvl.daily$water_year, sum)
TNloadings <- tapply(wvl.daily$TNLoad, wvl.daily$water_year, sum)

## spring
spring <- wvl.daily$Month == "Mar" |
  wvl.daily$Month == "Apr" |
  wvl.daily$Month == "May" |
  wvl.daily$Month == "Jun" |
  wvl.daily$Month == "Jul"

TPL_spring <- tapply(wvl.daily$TPLoad[spring],
```

```

        wvl.daily$water_year[spring], sum)
SRPL_spring <- tapply(wvl.daily$SRPLoad[spring],
                     wvl.daily$water_year[spring], sum)
TNL_spring <- tapply(wvl.daily$TNLoad[spring], wvl.daily$water_year[spring],
                     sum)

TPfrct <- TPL_spring/TPloadings
SRPfrct <- SRPL_spring/SRPloadings
TNfrct <- TNL_spring/TNloadings

TPLoads <- data.frame(Loadings=c(TPL_spring, TPloadings[-44])/1000,
                     years=rep(as.numeric(names(TPL_spring)), 2),
                     type=rep(c("Spring","Annual"), each=length(TPL_spring)),
                     frct=TPL_spring/TPloadings)

```

Warning in data.frame(Loadings = c(TPL_spring, TPloadings[-44])/1000, years = rep(as.numeric(names(TPL_spring)), : row names were found from a short variable and have been discarded

```

SRPLoads <- data.frame(Loadings=c(SRPL_spring, SRPloadings[-44])/1000,
                     years=rep(as.numeric(names(SRPL_spring)), 2),
                     type=rep(c("Spring","Annual"), each=length(SRPL_spring)),
                     frct=SRPL_spring/SRPloadings)

```

Warning in data.frame(Loadings = c(SRPL_spring, SRPloadings[-44])/1000, : row names were found from a short variable and have been discarded

```

#Frcts <- data.frame(frct=c(TPL_spring/TPloadings, severe),
#                   index=c(rep("Fraction", length(TPL_spring)),
#                           rep("Index", length(severe))),
#                   years=c(as.numeric(names(TPL_spring)), ))

fig1a <- xyplot(Loadings ~ years,
               panel=function(x,y, ...){
                 panel.xyplot(x, y, cex=0.5, ...)
                 panel.text(min(x), max(y), "(a)", adj = c(0, 1),
                           cex=0.7)
                 panel.abline(h=860, col="gray")
               },
               data=TPLoads, group=type, type="b",

```

```

      xlab=list(label="Water Year", cex=0.6),
      ylab=list(label="TP Loading (1000 kg)", cex=0.75),
      scales=list(cex=0.7),
      par.setting=list(fontsize=list(text=10)))#,
##      key=simpleKey(levels(as.factor(TPLoads$type)),
##                    lines=T, points=F, corner=c(0,1),
##                    x=0.1, y=0.95, columns=1, cex=0.6))

fig1b <- xyplot(frct~years,
                data=TPLoads[TPLoads$type=="Spring",],
                panel=function(x, y, ...){
                  panel.xyplot(x, y, cex=0.5, ...)
                  panel.loess(x, y, span=0.65, ...)
                  panel.text(min(x), max(y), "(c)", adj = c(0, 1),
                             cex=0.7)},
                scales=list(cex=0.7),
                par.setting=list(fontsize=list(text=10)),
                xlab=list(label="Water Year", cex=0.6),
                ylab=list(label="Spring TP loading fraction", cex=0.75)
                )

fig1c <- xyplot(Loadings ~ years,
                panel=function(x,y, ...){
                  panel.xyplot(x, y, cex=0.5, ...)
                  panel.text(min(x), max(y), "(b)", adj = c(0, 1),
                             cex=0.7)
                  panel.abline(h=186, col="gray")
                },
                data=SRPLoads, group=type, type="b",
                xlab=list(label="Water Year", cex=0.6),
                ylab=list(label="SRP Loading (1000 kg)", cex=0.75),
                scales=list(cex=0.7),
                par.setting=list(fontsize=list(text=10)),
                key=simpleKey(levels(as.factor(TPLoads$type)),
                              lines=T, points=F, corner=c(0,1),
                              x=0.025, y=0.85, columns=1, cex=0.6))

fig1d <- xyplot(frct~years,
                data=SRPLoads[SRPLoads$type=="Spring",],
                panel=function(x, y, ...){
                  panel.xyplot(x, y, cex=0.5, ...)
                  panel.loess(x, y, span=0.65, ...)

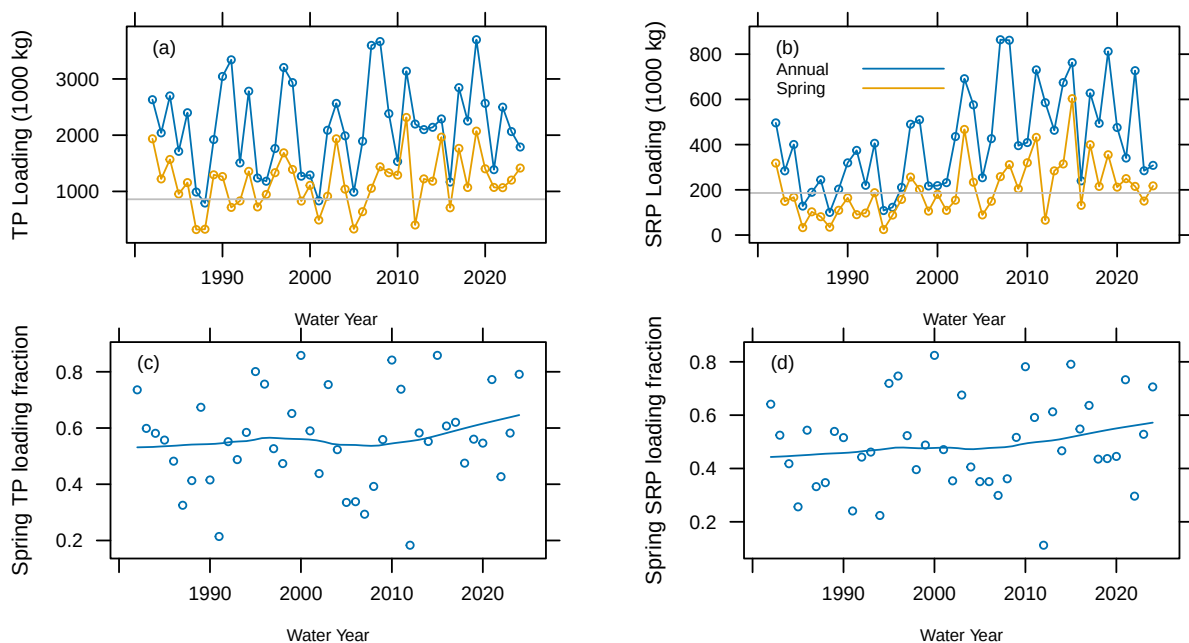
```

```

    panel.text(min(x), max(y), "(d)", adj = c(0, 1),
               cex=0.7)},
    scales=list(cex=0.7),
    par.setting=list(fontsize=list(text=10)),
    xlab=list(label="Water Year", cex=0.6),
    ylab=list(label="Spring SRP loading fraction", cex=0.75)
)

##tikz(file="TPLoadings_f1.tex", height=4.25, width=7.25,
##      standAlone = F)
print(fig1a, position=c(0,0.425,0.5,1), more=T)
print(fig1b, position=c(0,0,0.5,0.575), more=T)
print(fig1c, position=c(0.5,0.425,1,1), more=T)
print(fig1d, position=c(0.5,0,1,0.575), more=F)

```



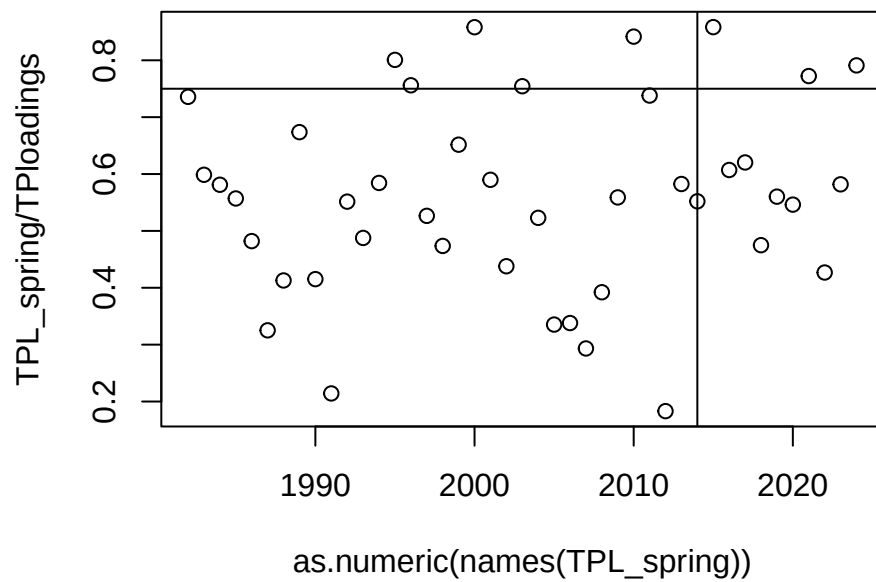
```
##dev.off()
```

Other figures not used

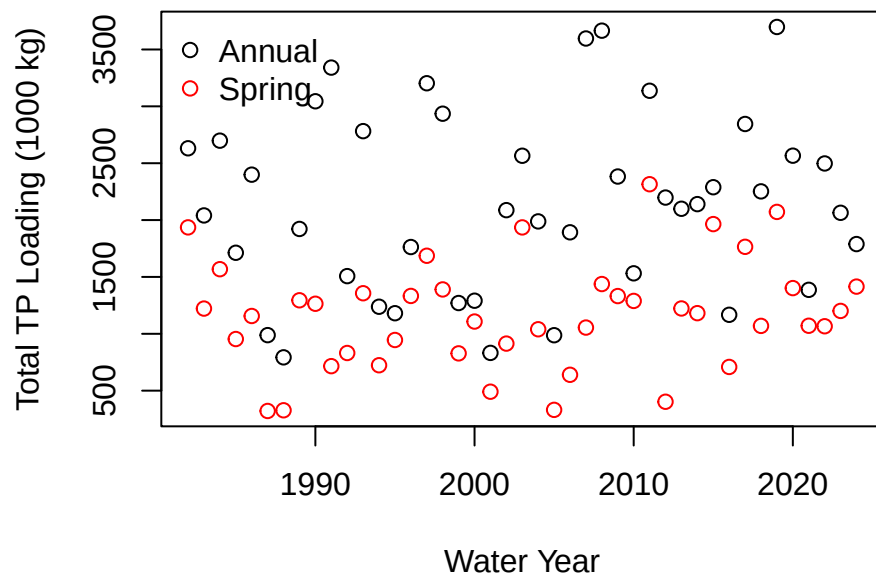
```

plot(as.numeric(names(TPL_spring)), TPL_spring / TPloadings)
abline(h=0.75)
abline(v=2014)

```



```
plot(as.numeric(names(TPL_spring)), TPloadings/1000,
     ylim=range(c(TPL_spring, TPloadings))/1000,
     xlab="Water Year", ylab="Total TP Loading (1000 kg)")
points(as.numeric(names(TPL_spring)), TPL_spring/1000, col="red")
legend("topleft", pch=c(1,1), col=c("black", "red"),
      legend=c("Annual", "Spring"), bty="n")
```



Fitting STL and Plotting

- Illustrating the edge effect

When using the flow data from January 1982 (the starting date of the water quality data), a few high flow events misrepresented the trend in the 1980s. The extreme high at the beginning the the trend is nothing but a small bump in the longer time series trend plot.

```
## Flow
f_heidelberg <- STL_Mplots_online (Var = "Flow", TSm = Flow_ts_monthly,
                                   TSd = Flow_ts_daily,
                                   aspect="xy",
                                   ylab=list(label="Flow (m3/s)", cex=0.8),
                                   xlab=list(label="Year", cex=0.8),
                                   axes.cex = 0.7, Ylim=c(75,210),
                                   ms.win=55, ss.deg=1, mt.win=135, tt.deg=1,
                                   ds.win=705, dt.win=3650, Log=F)$trend
f_usgs <- STL_Mplots_online (Var = "Flow_usgs", TSm = Flow_ts_monthly_usgs,
                             TSd = Flow_ts_daily_usgs,
                             aspect="xy",
                             ylab=list(label="Flow (m3/s)", cex=0.8),
                             xlab=list(label="Year", cex=0.8),
                             axes.cex = 0.7, Ylim=c(75,210),
                             ms.win=55, ss.deg=1, mt.win=135, tt.deg=1,
                             ds.win=705, dt.win=3650, Log=F)$trend

##tikz(paste(plotDIR, "fig_edge.tex", sep="/"),
##      height=5, width=7, standAlone = F)
print(f_usgs, position=c(0, 0.425, 1, 1), more = T)
print(f_heidelberg, position=c(0.385, 0, 0.975, 0.55), more =F)
```

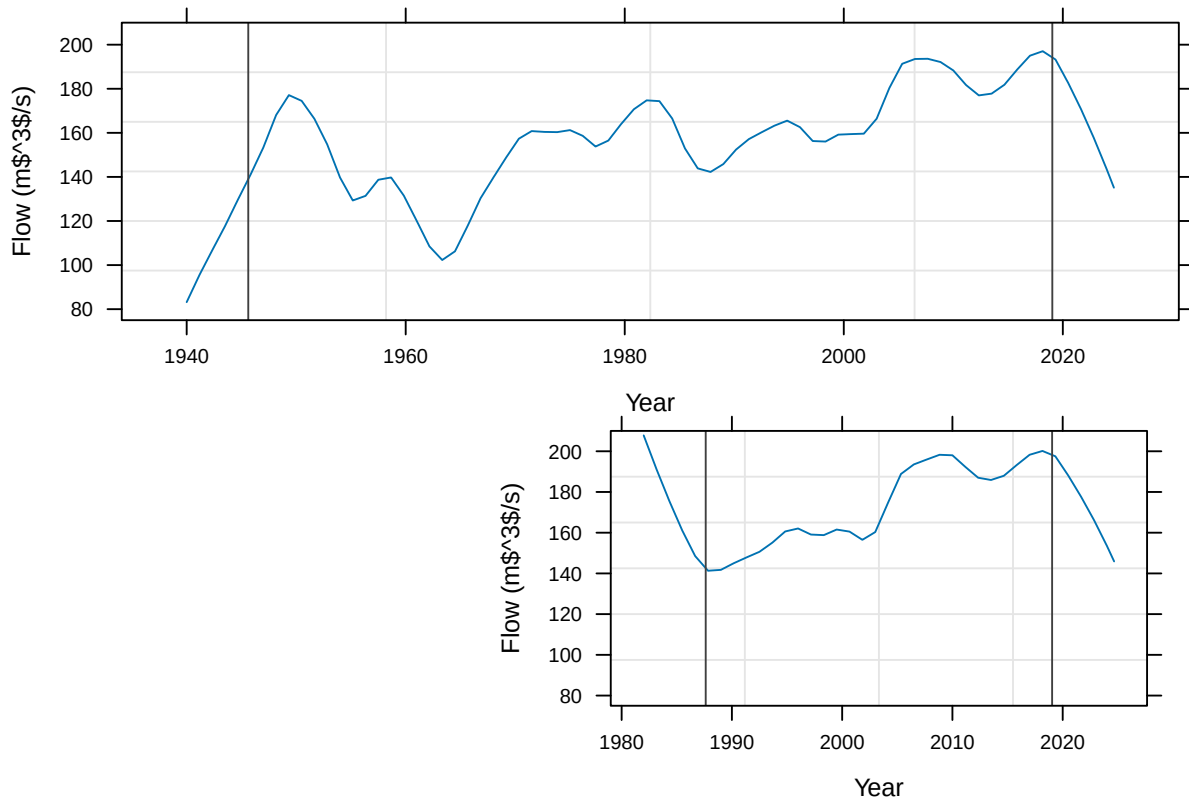


Figure 1: A comparison of the STL estimated discharge long-term trend using the full USGS record (from 1940 to 2024, upper panel) and the same using the Heidelberg flow record started in 1982 (lower panel). The initial high in the lower panel near the early 1980s does not exist in the trend based on the full record.

```
##dev.off()
```

- Preparing for Figures 4-8

```
ms_tploads <- STL_Mplots_online (Var = "TPLoad", TSm = TPLoad_ts_monthly,
                                TSd = TPLoad_ts_daily, aspect="xy",
                                ylab=list(label="TP Loading (kg/day)", cex=0.8),
                                xlab=list(label="Year", cex=0.8),
                                axes.cex = 0.7, Log=F,
                                ms.win=55, ss.deg=1,
                                mt.win=135, tt.deg=1, ds.win=705, dt.win=3650)

ms_srploads <- STL_Mplots_online (Var = "SRPLoad", TSm = SRPLoad_ts_monthly,
                                TSd = SRPLoad_ts_daily, aspect="xy",
```

```

        ylab=list(label="SRP Loading (kg/day)", cex=0.8),
        xlab=list(label="Year", cex=0.8),
        axes.cex = 0.7,
        ms.win=55, ss.deg=1,
        mt.win=135, tt.deg=1, ds.win=705, dt.win=3650,
        Log=F)

ms_tnloads <- STL_Mplots_online (Var = "TNLoad", TSm = TNLoad_ts_monthly,
                                TSd = TNLoad_ts_daily, aspect="xy",
                                ylab=list(label="TN Loading (kg/day)", cex=0.8),
                                xlab=list(label="Year", cex=0.8),
                                axes.cex = 0.7,
                                ms.win=55, ss.deg=1,
                                mt.win=135, tt.deg=1, ds.win=705, dt.win=3650,
                                Log=F)

ms_tnconc <- STL_Mplots_online (Var = "TN", TSm = TN_ts_monthly,
                                TSd = TN_ts_daily, aspect="xy",
                                ylab=list(label="TN (mg/L)", cex=0.8),
                                xlab=list(label="Year", cex=0.8),
                                axes.cex = 0.7,
                                ms.win=55, ss.deg=1,
                                mt.win=135, tt.deg=1, ds.win=705, dt.win=3650,
                                Log=F)

ms_tpconc <- STL_Mplots_online (Var = "TP", TSm = TP_ts_monthly,
                                TSd = TP_ts_daily, aspect="xy",
                                ylab=list(label="TP (mg/L)", cex=0.8),
                                xlab=list(label="Year", cex=0.8),
                                axes.cex = 0.7,
                                ms.win=55, ss.deg=1,
                                mt.win=135, tt.deg=1, ds.win=705, dt.win=3650,
                                Log=F)

ms_srpconc <- STL_Mplots_online (Var = "SRP", TSm = SRP_ts_monthly,
                                TSd = SRP_ts_daily, aspect="xy",
                                ylab=list(label="SRP (mg/L)", cex=0.8),
                                xlab=list(label="Year", cex=0.8),
                                axes.cex = 0.7,
                                ms.win=55, ss.deg=1,
                                mt.win=135, tt.deg=1, ds.win=705, dt.win=3650,
                                Log=F)

```



```

ms_flow <- STL_Mplots_online (Var = "Flow_usgs",
                             TSm = Flow_ts_monthly_usgs,
                             TSd = Flow_ts_daily_usgs,
                             aspect="xy",
                             ylab=list(label="Flow (m3/s)", cex=0.8),
                             xlab=list(label="Year", cex=0.8),
                             axes.cex = 0.7,
                             ms.win=55, ss.deg=1, mt.win=135, tt.deg=1,
                             ds.win=705, dt.win=3650, Log=F)

ms_precip <- STL_Mplots_online (Var = "Precipitation", TSm = TS_precip_monthly2,
                                TSd = TS_precip_daily,
                                aspect="xy",
                                ylab=list(label="Precip (mm/day)", cex=0.8),
                                xlab=list(label="Year", cex=0.8),
                                axes.cex = 0.7,
                                ms.win=55, ss.deg=1, mt.win=135, tt.deg=1,
                                ds.win=705, dt.win=3650, Log=F)

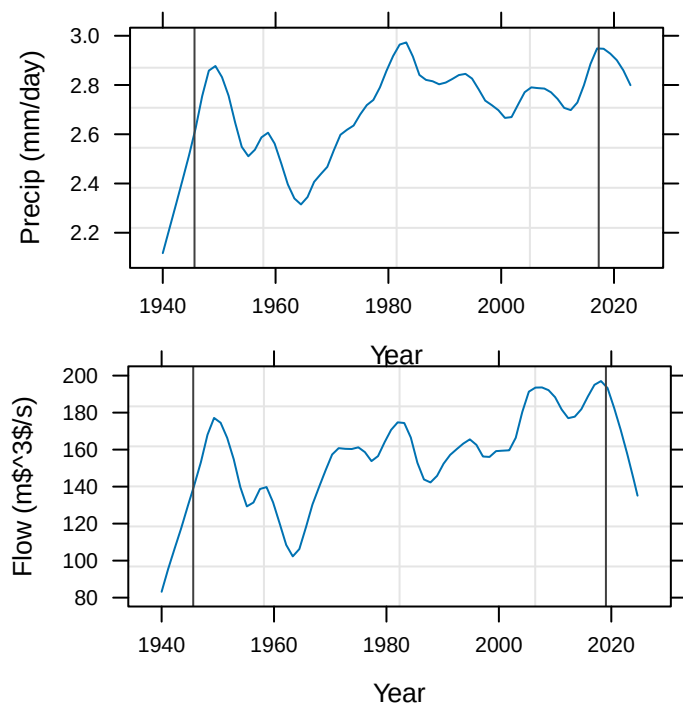
```

- Figure 4 flow & precip trends

```

##      tikz(paste(plotDIR, file="est_f2.tex", sep="/"),
##      height=4.25, width=4, standalone = F)
print(ms_precip$trend, position = c(0.01, 0.415, 0.99, 1), more=T)
print(ms_flow$trend, position=c(0, 0, 1, 0.585), more=F)

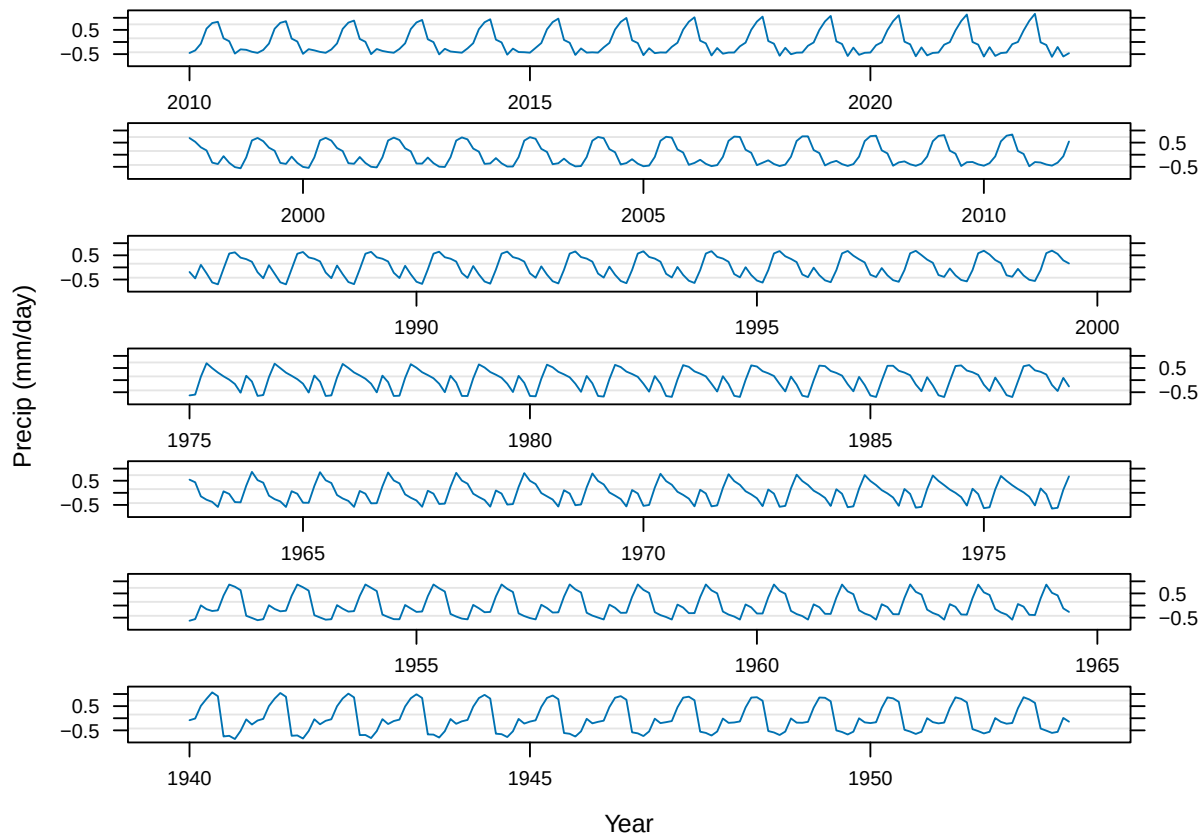
```



```
## dev.off()
```

- Figure 5

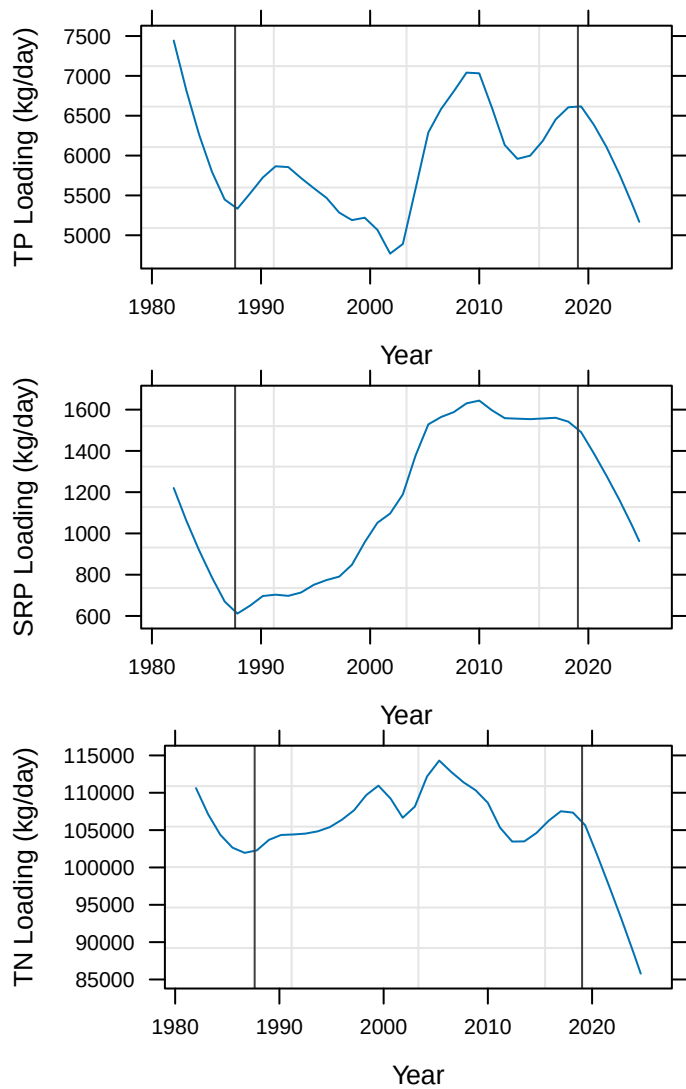
```
## Figure 5: precip ocs
## tikz(paste(plotDIR, "est_f4.tex", sep="/"),
##       height=5.75, width=7, standalone=F)
print(ms_precip$seasonosc)
```



```
## dev.off()
```

- Figure 6 TP & SRP & TN Loading trends

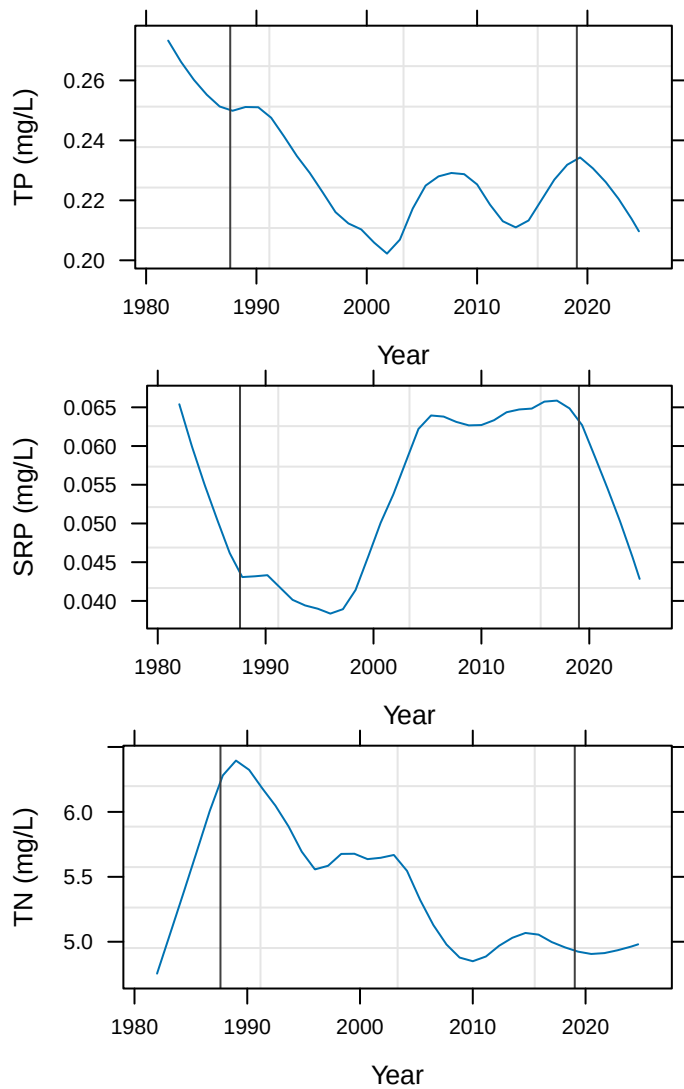
```
## tikz(paste(plotDIR, file="est_f3.tex", sep="/"),
##      height=6.25, width=4, standalone = F)
print(ms_tploads$trend, position=c(0, 0.6, 1, 1), more=T)
print(ms_srploads$trend, position=c(0, 0.3, 1, 0.7), more=T)
print(ms_tnloads$trend, position=c(0, 0, 1, 0.4), more=F)
```



```
## dev.off()
```

- Figure 7 TP, SRP, TN concentrations

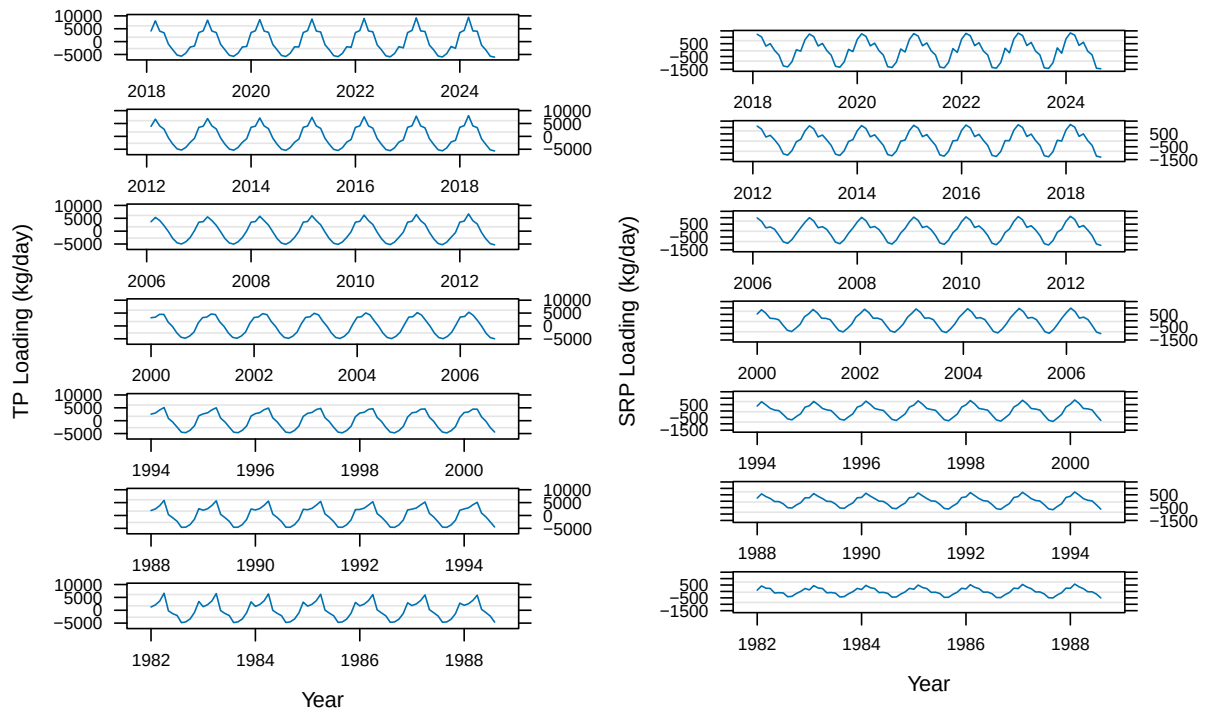
```
## tikz(paste(plotDIR, "est_conc.tex", sep="/"),
##      height=6.25, width=4, standalone=F)
print(ms_tpconc$trend, position=c(0, 0.6, 1, 1), more=T)
print(ms_srpconc$trend, position=c(0, 0.3, 1, 0.7), more=T)
print(ms_tnconc$trend, position=c(0, 0, 1, 0.4), more=F)
```



```
## dev.off()
```

- Figure 8: TP/SRP ocs

```
## tikz(paste(plotDIR, "est_f5.tex", sep="/"),
##      height=5.75, width=8, standalone=F)
print(ms_tploads$seasonosc, position=c(0, 0, 0.515, 1), more=T)
print(ms_srploads$seasonosc, position=c(0.485, 0, 1, 1), more=F)
```



```
## dev.off()
```

End of Part III